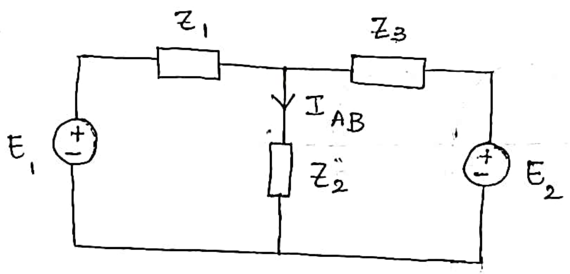
MODULE - 2

NETWORK THEOREMS.

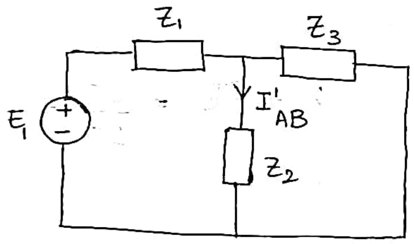
(1) Superposition theorem :

It states that "In a linear bilateral circuit with several sources, the current & voltage for any element in the circuit is the sum of the currents & voltages produced by each source acting independently."

Eg:-

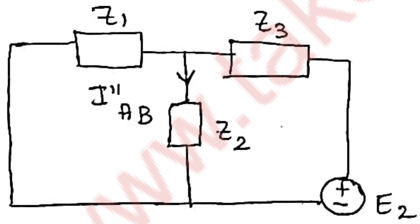


Case 1 : According to superposition theorem, consider each source at a time i.e, E_1 voltage source & E_2 will be short circuited.



Using any of the network reduction techniques find I'_{AB}

Case 2: Now consider source E_2 acting alone & short circuit E_1



Again use any network reduction techniques find I''_{AB}

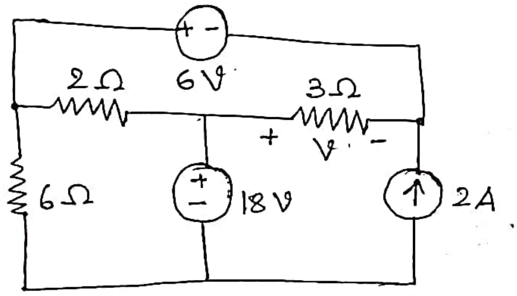
∴ According to superposition theorem

$$I = I'_{AB} + I''_{AB}$$

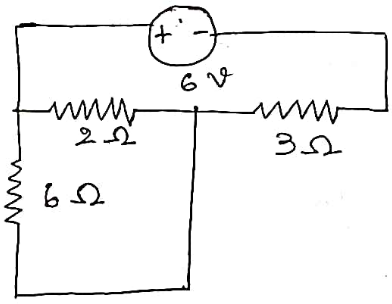
Note :-

(1) Voltage source must be replaced by short circuit
(2) Current source must be replaced by open circuit.
(3) If there are dependent current and voltage sources present in circuit then it should not be replaced keep as it is.

(1) Find voltage 'V' across $3\,\Omega$ resistor.



Keep 6V source, short circuit the 18V source & open circuit the 2A current source.



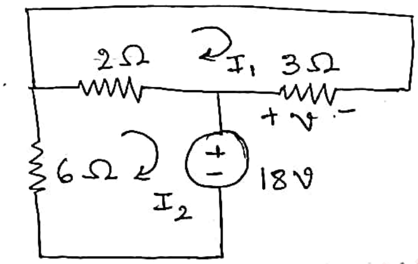
$6\,\Omega \parallel 2\,\Omega$

$\therefore \; R_P = 1.5\,\Omega$.

$I_T = \dfrac{V}{R_T} = \dfrac{6}{1.5+3} = 1.333A$.

$V' = 3I_T = \underline{4V}$.

Consider 18V, short 6V, open 2A.



$-3I_1 - 2(I_1 - I_2) = 0$

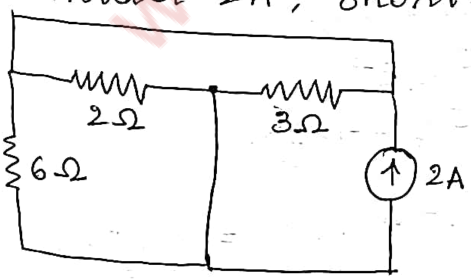
$-18 - 6I_2 - 2(I_2 - I_1) = 0$.

$5I_1 - 2I_2 = 0$. $\qquad I_1 = -1A$

$8I_2 - 2I_1 = -18$. $\qquad I_2 = -2.5A$.

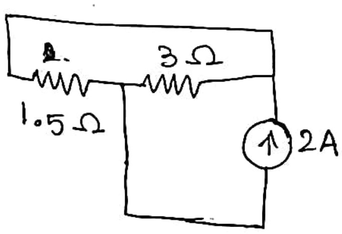
$V'' = 3I_1 = 3 \times 1 = \underline{3V}$.

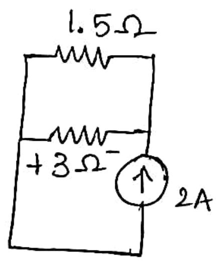
Consider 2A, short 6V and 18V.



$I = \dfrac{2(1.5)}{1.5+3} = 0.666A$

$V''' = -3I = \underline{-2V}$.

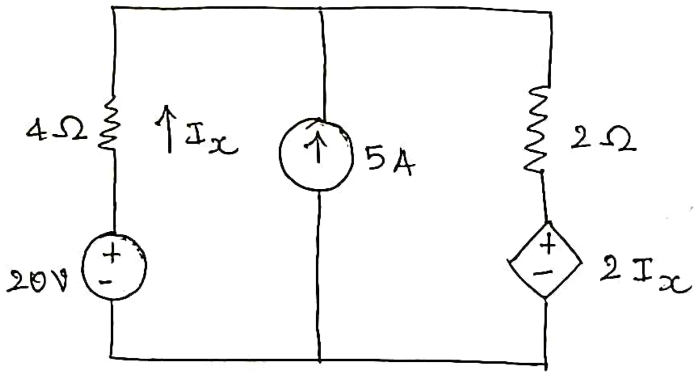




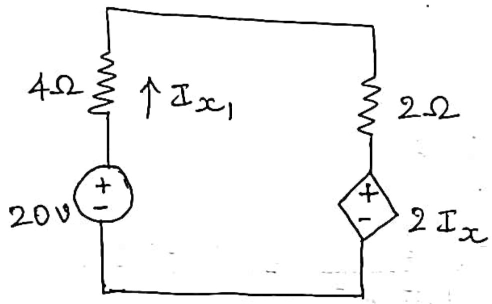
$\therefore \; V = V' + V'' + V'''$

$V = 4 + 3 - 2 = \underline{\underline{5V}}$.

(2) Find the current I_x



consider 20V, ~~sho~~ open 5A.



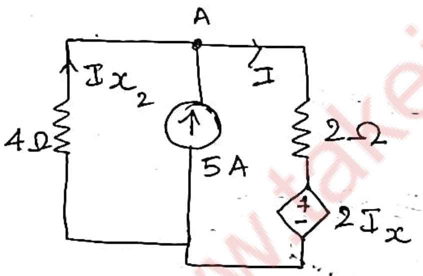
Apply KVL

$$20 - 6I_{x_1} - 2I_{x_1} = 0.$$

$$20 - 8I_{x_1} = 0.$$

$$I_{x_1} = \underline{\underline{2.5A}}.$$

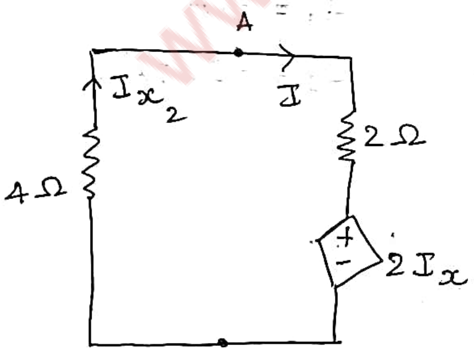
consider 5A, short 20V.



consider node A,

$$I_{x_2} + 5 = I. \quad \text{— (1)}$$

Apply KVL,



$$-4I_{x_2} - 2I - 2I_{x_2} = 0.$$

$$-6I_{x_2} - 2I = 0.$$

$$3I_{x_2} + \cancel{I} = 0. \quad \text{— (2)}$$

$$I_{x_2} - \cancel{I} = -5$$

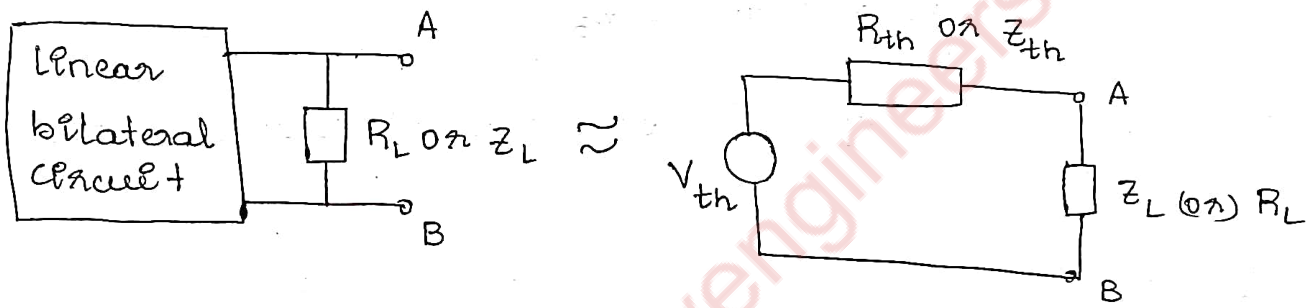
$$4I_{x_2} = -5$$

$$I_{x_2} = \underline{\underline{-1.25A}}.$$

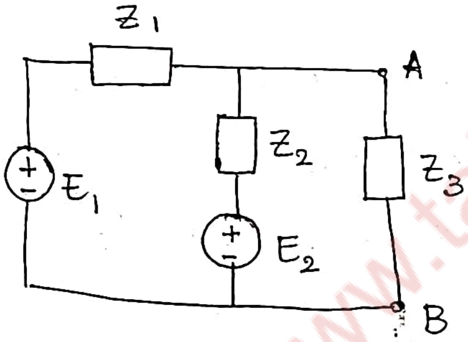
$$\therefore I_x = I_{x_1} + I_{x_2} = 2.5 - 1.25 = \underline{\underline{1.25A}}$$

* Thevenins theorem

Any linear bilateral circuit containing several voltage and resistances (or impedances) can be replaced into an equivalent two terminal ~~connected~~ ~~to~~ a ~~load~~ circuit with a single voltage source in series with a resistance (or impedance) connected to a load.



<u>Proof</u> :- Consider the below circuit.



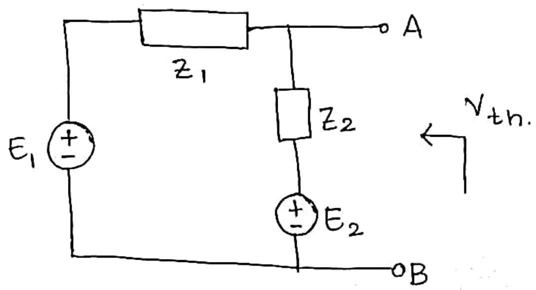
First apply any network simplification technique and find load current. Let us apply Mesh analysis.

$$\begin{bmatrix} Z_1+Z_2 & -Z_2 \\ -Z_2 & Z_2+Z_3 \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \end{bmatrix} = \begin{bmatrix} E_1 - E_2 \\ E_2 \end{bmatrix}$$

$$I_2 = \frac{\begin{bmatrix} Z_1+Z_2 & E_1-E_2 \\ -Z_2 & E_2 \end{bmatrix}}{\Delta} = \frac{E_2(Z_1+Z_2) + Z_2(E_1-E_2)}{(Z_1+Z_2)(Z_2+Z_3) - Z_2^2}$$

$$I_2 = \frac{Z_1 E_2 + Z_2 E_1}{Z_1 Z_2 + Z_2 Z_3 + Z_1 Z_3} \quad \text{—— ①.}$$

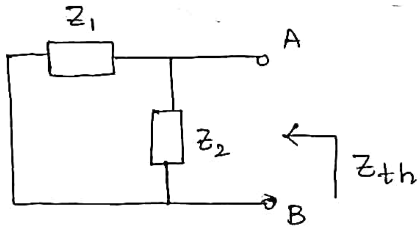
Now let us apply thevenin's theorem and find load current.



$$V_{AB} = V_{th} = I Z_2 + E_2.$$

$$\text{But } I = \frac{E_1 - E_2}{Z_1 + Z_2}$$

$$V_{AB} = \frac{(E_1 - E_2) Z_2}{Z_1 + Z_2} + E_2.$$



$$Z_{th} = \frac{Z_1 Z_2}{Z_1 + Z_2}$$

$$\therefore \; I_L = \frac{V_{th}}{Z_L + Z_{th}} = \frac{\dfrac{(E_1 - E_2) Z_2}{Z_1 + Z_2} + E_2}{Z_3 + \dfrac{Z_1 Z_2}{Z_1 + Z_2}}$$

$$I_L = \frac{Z_1 E_2 + Z_2 E_1}{Z_1 Z_2 + Z_2 Z_3 + Z_1 Z_3} \quad \text{—— ②.}$$

comparing eqn ① and ② load current eqn is same.

★ **Limitations.**

(1) This theorem cannot be applied to circuit with non-linear elements.

(2) There should not be magnetic coupling b/w the loads

(3) cannot be applied to unilateral networks

(4) There should not be any controlled sources with load.

★ **Steps to solve**

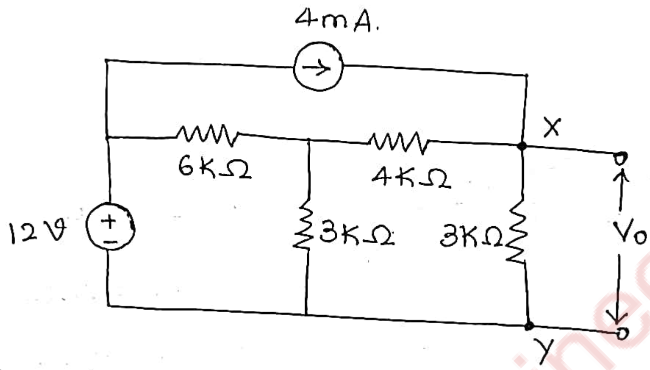
(1) Remove the load through which current is required to be calculated.

(2) Calculate the thevenin's equivalent voltage V_{th} using any network simplification techniques.

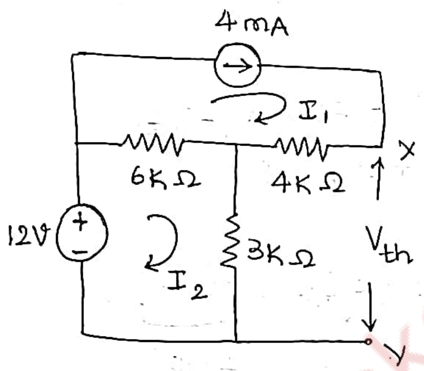
(3) Calculate the equivalent impedance Z_{eq}, by replacing voltage source by short circuit and current source by open circuit.

(4) Now using V_{th} and Z_{eq} calculate the current through the given load $I = \dfrac{V_{th}}{Z_L + Z_{eq}}$.

(1) Obtain the thevenin's equivalent of network b/w terminals x and Y. Also find V_o.



$Z_L = 3K\Omega$, Remove the load and find V_{th}



$I_1 = 4mA.$

Apply KVL,

$$12 - 6K(I_2 - 4\times10^{-3}) - 3\times10^3(I_2) = 0.$$

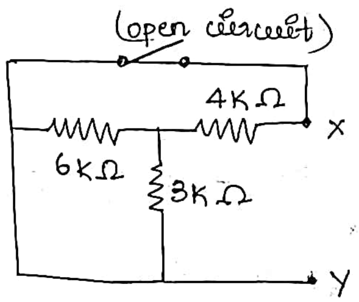
$$12 - 6\times10^3 I_2 + 24 - 3\times10^3 I_2 = 0.$$

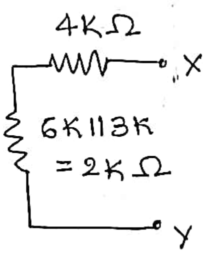
$$9\times10^3 I_2 = 36.$$

$$I_2 = 4mA.$$

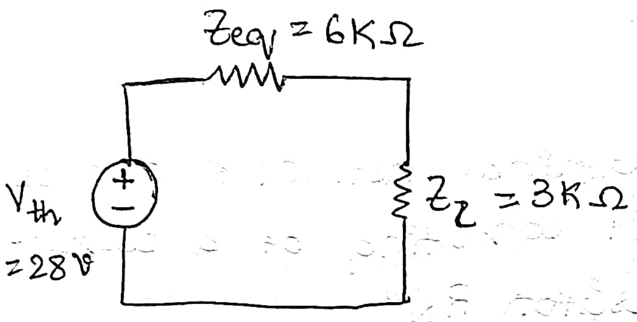
$$V_{th} = V_{3K} + V_{4K} = 12V + 16V = 28V.$$

to find equivalent impedance Z_{eq}.





$$\therefore Z_{eq} = 4K + 2K = 6K\Omega.$$



$$I = \frac{V_{th}}{Z_L + Z_{eq}} = \frac{28}{3K + 6K} = 0.00311$$

$$I = 3.11 mA$$

Note :-

If the given network consists of dependent sources.

(1) Calculate V_{th} by any of the network simplification methods.

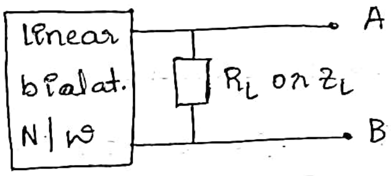
(2) Then short the terminals & obtain current through it say I_{sc} (note : none of the source should be replaced by short or open circuit)

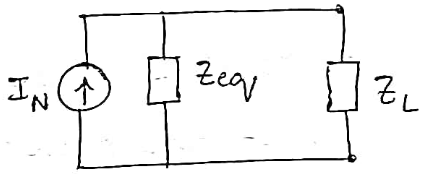
$$\therefore \quad Z_{eq} = \frac{V_{oc}}{I_{sc}}$$

$$V_{oc} = V_{th}.$$

* Norton's theorem

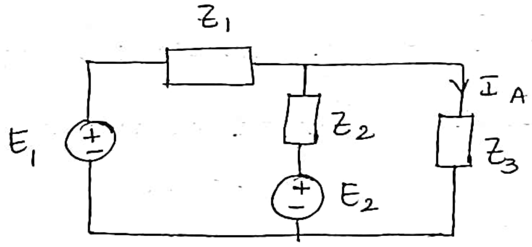
"A linear & bidirectional two-terminal network can be replaced by an equivalent circuit consisting of a current source I_N in parallel with a resistor R_N"





where $I_L = \dfrac{I_N \times Z_{eq}}{Z_{eq} + Z_L}$

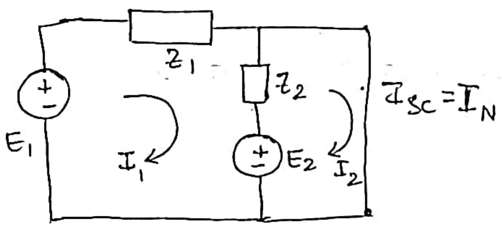
Proof:



By applying mesh analysis

$$I_L = \frac{Z_1 E_2 + Z_2 E_1}{Z_1 Z_2 + Z_3 Z_2 + Z_1 Z_3} \quad \text{--- (1)}$$

Now to calculate I_N. Replace load by a short circuit, hence I_N is called short circuit current.

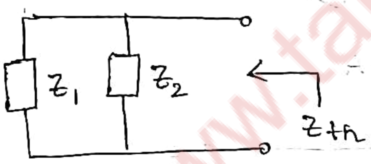


Apply mesh analysis

$$\begin{bmatrix} Z_1 + Z_2 & (-Z_2) \\ (-Z_2) & (Z_2) \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \end{bmatrix} = \begin{bmatrix} E_1 - E_2 \\ E_2 \end{bmatrix}$$

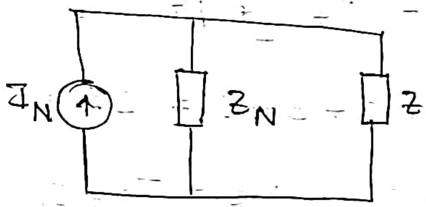
$$I_2 = I_L = I_{sc} = I_N = \frac{\begin{vmatrix} Z_1 + Z_2 & E_1 - E_2 \\ -Z_2 & E_2 \end{vmatrix}}{\Delta} = \frac{Z_1 E_2 + Z_2 E_1}{Z_1 Z_2}$$

Find Z_{eq} (same as Z_{th})



$$Z_{th} = \frac{Z_1 Z_2}{Z_1 + Z_2} = Z_{eq} = Z_N = Z.$$

Now Norton's equivalent is



$$I_L = \frac{I_N \times Z_{th}}{Z_L + Z_{th}}.$$

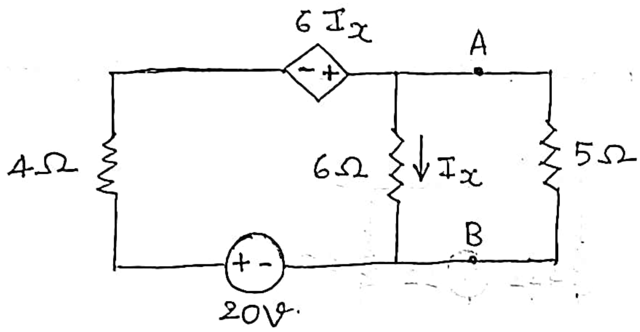
$$I_L = \frac{Z_1 E_2 + Z_2 E_1}{Z_1 Z_2 + Z_2 Z_3 + Z_1 Z_3} \quad \text{--- (2)}.$$

Equ^n (2) is same as equ^n (1), hence proved.

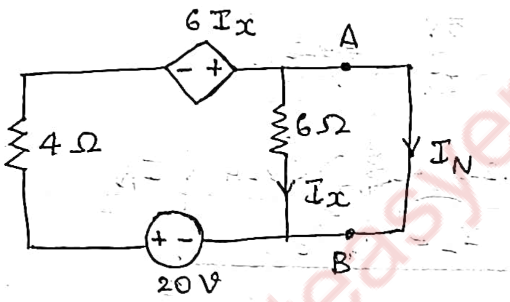
I_N is same as V_{th} but in norton's theorem we short circuit the load.

R_N is same R_{th} $\left(R_N = \dfrac{V_{oc}}{I_{sc} \text{ or } I_N} \text{ for dependent source} \right)$

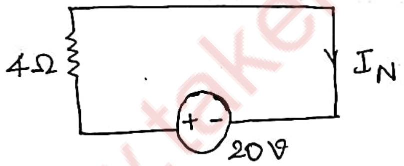
(1) Determine the Norton's equivalent circuit across AB terminals in the network. Hence determine current in 5Ω resistor



To find I_N short AB terminals.

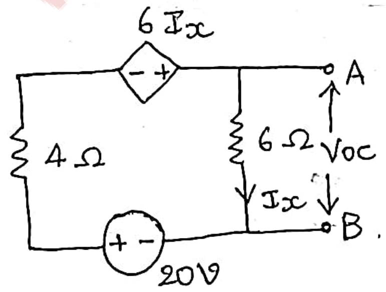


current through 6Ω is 0
hence $6I_x$ is also 0
(because current tends to flow in least resistance path).



$$I_N = \frac{20}{4} = 5A.$$

To find R_N or $R_{eq} = \dfrac{V_{oc}}{I_N}$ (due to dependent source)



Apply KVL,

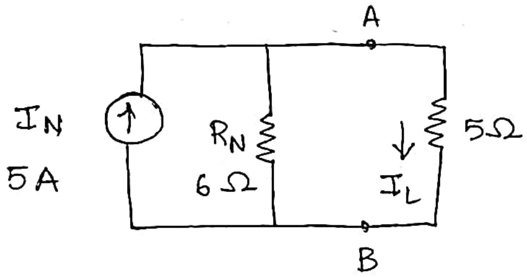
$$20 - 4I_x + 6I_x - 6I_x = 0.$$

$$20 = 4I_x$$

$$I_x = 5A.$$

$$V_{oc} = 6I_x = 30V.$$

$$R_{eq} = \frac{V_{oc}}{I_N} = \frac{30}{5} = 6\Omega.$$

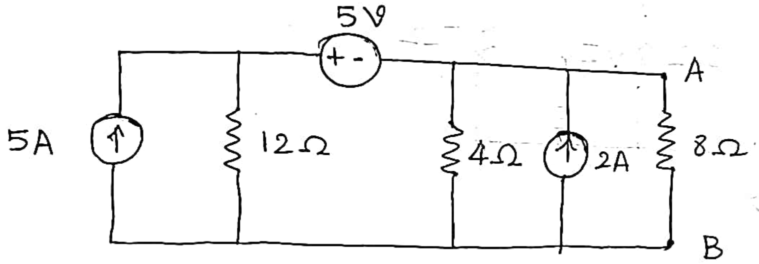


$$I_L = \frac{I_N \times 6}{6+5}$$

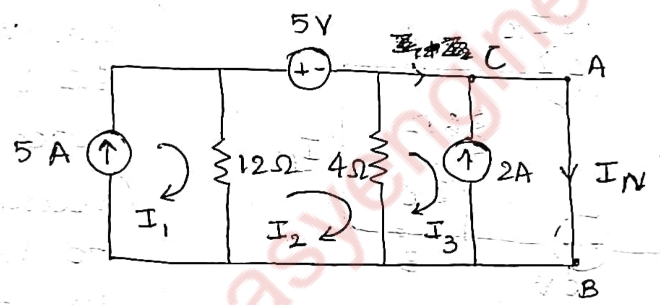
$$= \frac{5 \times 6}{6+5}$$

$$I_L = 2.7272A.$$

(2) Find the current through 8 Ω resistance by using



to find I_N.



$I_1 = 5A.$

Apply KCL at node c.

$$I_3 + 2 = I_N.$$

Apply KVL,

$$-12(I_2 - I_1) - 5 - 4(I_2 - I_3) = 0.$$

$$-16 I_2 + 55 + 4 I_3 = 0.$$

$$-4(I_3 - I_2) = 0.$$

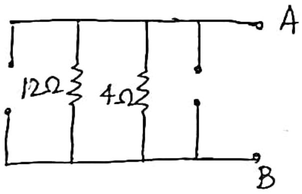
$$-4I_3 + 4I_2 = 0.$$

$$I_2 = 4.5833A.$$

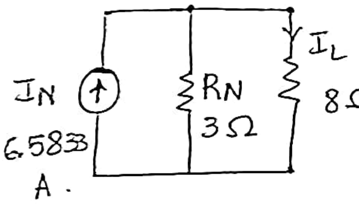
$$I_3 = 4.5833A.$$

$$I_N = 6.5833A.$$

To find R_N,



$$R_N = \frac{12 \times 4}{12+4} = 3\Omega$$



$$I_L = I_N \cdot \frac{R_N}{R_L + R_N}$$

$$= \frac{6.5833 \times 3}{8+3}$$

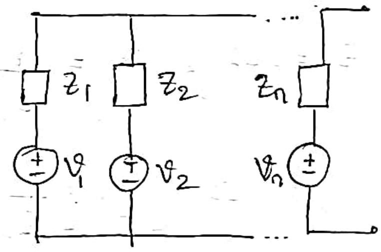
$$I_L = 1.7954 A.$$

* Millman's theorem.

It is possible to combine number of voltage sources or current sources into a single equivalent V/g. & c/t source. i.e, if n V/g sources $v_1, v_2, v_3, \ldots v_n$ having internal impedances $z_1, z_2, \ldots z_n$ are in parallel as shown in fig below, then these source impedance Z_m where V_M & Z_M are given by.

$$V_M = \frac{v_1 Y_1 + v_2 Y_2 + \ldots + v_n Y_n}{Y_1 + Y_2 + \ldots + Y_n}$$

$$Z_M = \frac{1}{Y_1 + Y_2 + \ldots Y_n}.$$

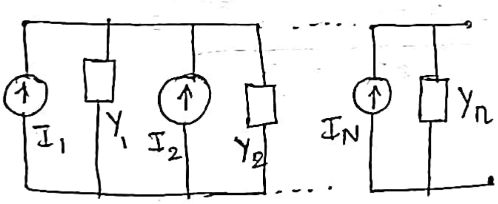


Proof: from the circuit diagram

$$I_1 = \frac{v_1}{Z_1} \quad \text{but} \quad Y_1 = \frac{1}{Z_1} \quad \therefore I_1 = v_1 Y_1$$

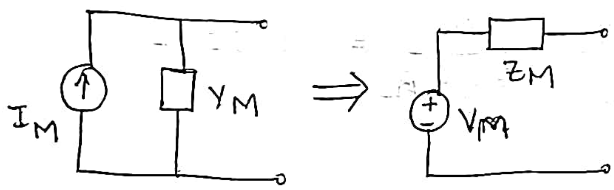
$|||^{ly}$ $I_2 = v_2 Y_2$, $I_3 = v_3 Y_3$ + ... $I_n = v_n Y_n$

converting voltage sources into ~~c/t~~ current sources



where $I_M = I_1 + I_2 + \ldots + I_n.$

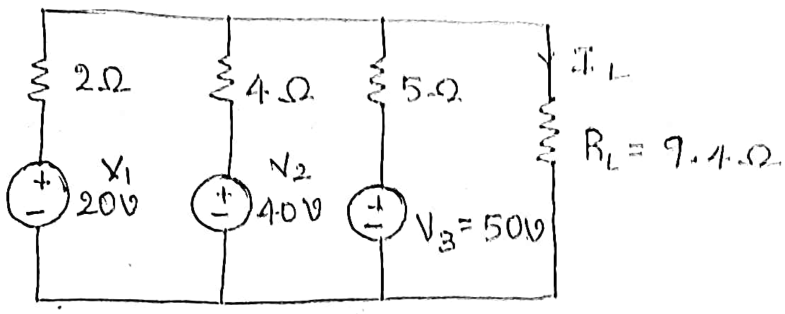
$Y_M = Y_1 + Y_2 + \ldots + Y_n$



where $V_M = \frac{I_M}{Y_M}$

$$Z_M = \frac{1}{Y_M} = \frac{1}{Y_1 + Y_2 + \ldots + Y_M}.$$

(1) Using millman's theorem, find I_L through R_L



$V_1 = 20V$, $V_2 = 40V$, $V_3 = 50V$.

$Z_1 = 2\Omega$, $Y_1 = \frac{1}{Z_1} = \frac{1}{2} ℧$

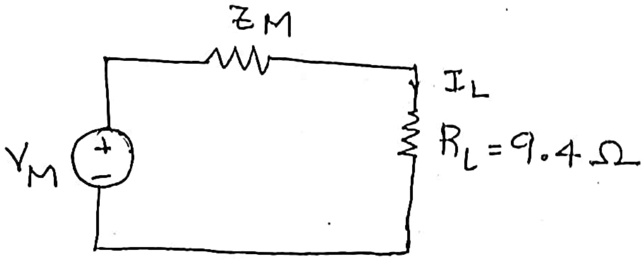
$Z_2 = 4\Omega$, $Y_2 = \frac{1}{Z_2} = \frac{1}{4} ℧$

$Z_3 = 5\Omega$, $Y_3 = \frac{1}{Z_3} = \frac{1}{5} ℧$

$$Z_M = \frac{1}{Y_1 + Y_2 + Y_3} = \frac{1}{1/2 + 1/4 + 1/5} = 1.0526\Omega$$

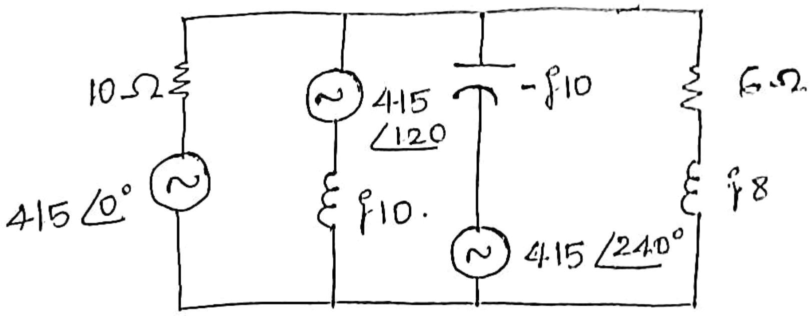
$$V_M = \frac{V_1Y_1 + V_2Y_2 + V_3Y_3}{Y_1 + Y_2 + Y_3} = \frac{20(1/2) + 40(1/4) + 50(1/5)}{0.95}$$

$V_M = 31.5789V$.



$$I_L = \frac{V_M}{Z_M + R_L} = 3.0211A.$$

(2) Find the current in $(6+j8)\,\Omega$ impedance.



$V_1 = 415\,V$, $V_2 = 415\angle 120°\,V$, $V_3 = 415\angle 240°\,V$.

$$Z_1 = Y_1 = \frac{1}{10} = 0.1\,\mho$$

$$Z_2 = Y_2 = \frac{1}{j10}\,\mho = 0.1\angle -90°\,\mho$$

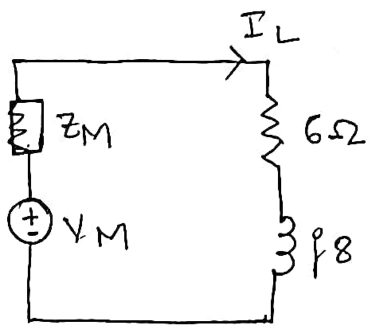
$$Z_3 = Y_3 = \frac{1}{-j10}\,\mho = 0.1\angle 90°\,\mho$$

$$Z_M = \frac{1}{Y_1 + Y_2 + Y_3} = \frac{1}{0.1 + 0.1\angle -90° + 0.1\angle 90°} = 10\,\Omega$$

$$V_M = \frac{V_1 Y_1 + V_2 Y_2 + V_3 Y_3}{Y_1 + Y_2 + Y_3}$$

$$= \frac{415(0.1) + 415\angle 120°(0.1\angle -90°) + 415\angle 240°(0.1\angle 90°)}{0.1}$$

$$V_M = 1133.80\,V$$



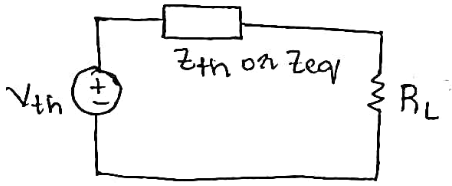
$$I_L = \frac{V_M}{Z_M + Z_L}$$

$$= \frac{1133.8}{10 + 6 + j8}$$

$$I_L = 63.38135\angle -26.56°$$

* Maximum power transfer theorem (DC circuits)

Statement: In active network, maximum power transfer to the load take place when $R_L = |Z_{eq}|$ i.e when load resistance is equal to Z_{eq} or Z_{th} (absolute magnitude of Z_{eq} or Z_{th}).



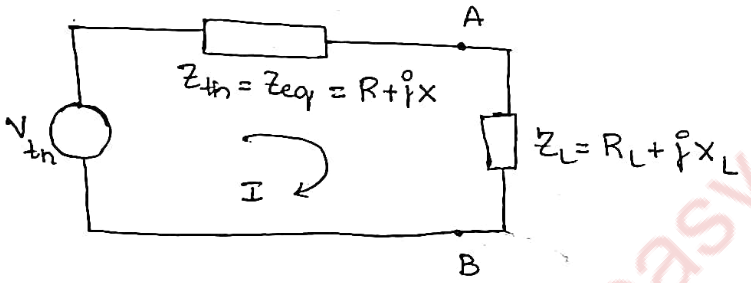
$$I_L = \frac{V_{th}}{Z_{eq} + Z_L}$$

$$P_{max} = I^2 R_L = \left[\frac{V_{th}}{Z_{eq} + R_L}\right]^2 R_L$$

* Maximum power transfer theorem for AC circuits.

In active network, max power transfer to the load takes place when the load impedance is the complex conjugate of an equivalent impedance of the network, i.e $Z_L = Z_{eq}^* = R - jX$

Proof :- consider a thevenin's equivalent circuit as shown below



W.K.T

$$I = \frac{V_{th}}{Z_L + Z_{eq}} = \frac{V_{th}}{R + jX + R_L + jX_L}$$

$$I = \frac{V_{th}}{(R+R_L) + j(X+X_L)}, \quad \text{then } P_L = I^2 R_L$$

$$P_L = \frac{V_{th}^2}{(R+R_L)^2 + j(X+X_L)^2} \cdot R_L \quad \text{— (1).}$$

To find maximum power transfered, according to maximum theorem.

$$\frac{\partial P_L}{\partial X_L} = 0 \Rightarrow \frac{0 - V_{th}^2 R_L \left(2(X+X_L)\right)}{\left[(R+R_L)^2 + (X+X_L)^2\right]^2} = 0.$$

$$X + X_L = 0 \quad (\text{or}) \quad X_L = -X$$

also $\frac{\partial P_L}{\partial R_L} = 0$ & substituting $X_L = -X$ in eqn (1).

$$\frac{\partial}{\partial R_L}\left[\frac{V_{th}^2 R_L}{(R+R_L)}\right] = 0.$$

$$\frac{(R+R_L)^2 V_{th}^2 - V_{th}^2 R_L \cdot 2(R+R_L)}{[(R+R_L)^2]^2} = 0.$$

simplifying, take V_{th}^2 as common & make it zero

then $(R+R_L)^2 - 2R_L(R+R_L) = 0.$

$$R^2 + R_L^2 + 2RR_L - 2R_LR - 2R_L^2 = 0.$$

$$R^2 - R_L^2 = 0 \Rightarrow R_L^2 = R^2 \text{ or } R_L = R.$$

Thus $R_L + jX_L = R - jX$

(or) $Z_L = Z_{eq}^* \Rightarrow$ Hence proved.

$$\therefore P_{max} = I^2 R_L = \frac{V_{th}^2 R_L}{(R_L + R_{eq})^2} = \frac{V_{th}^2 R_L}{(R_L + R_L)^2} = \frac{V_{th}^2 R_L}{(2R_L)^2} = \frac{V_{th}^2}{4R_L}$$

But the efficiency of the circuit is 50% when $P = P_{max}$.